Kilometer-scale single-shot macroscopic Fourier ptychography with meteorology-aware restoration

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Abstract

This document provides supplementary information to "Kilometer-scale single-shot macroscopic Fourier ptychography with meteorology-aware restoration". To further clarify our kilometer-scale single-shot macroscopic Fourier ptychography (FP), additional experiments were conducted and reported.

1 The Kilometer-scale Single-shot Macroscopic Fourier Ptychography System

Conventional macroscopic Fourier ptychography (FP) generally requires either mechanical aperture scanning or sequential camera translation to acquire multiple sub-aperture measurements. Although such scanning-based configurations can synthesize a large effective aperture, they inevitably introduce several practical limitations in long-range experiments. First, the acquisition of sub-aperture images is time-consuming because different aperture positions are recorded sequentially. Second, mechanical motion requires accurate positioning and stable opto-mechanical alignment. Third, for kilometer-scale horizontal-path imaging, the atmospheric state may vary during the acquisition process, leading to temporal inconsistency among sub-aperture measurements. These factors may degrade the phase consistency required for Fourier-domain spectrum stitching and limit the applicability of scanning-based macroscopic FP under dynamic atmospheric conditions.

1.1 The 2 × 2 Camera Array

In this work, we implement a kilometer-scale single-shot macroscopic FP system based on a 2 × 2 camera-array architecture, as shown in Fig. S1. Instead of scanning a single receiving aperture, four spatially separated sub-aperture channels are used to simultaneously record the coherent images of the distant target within a single exposure. Each channel integrates a near-infrared camera, a long-focal-length imaging lens, a narrowband optical filter, and a polarizer. The customized receiving lens was designed as a compact long-focal-length module optimized for 1064 nm narrowband imaging. The compact form factor facilitates integration and alignment of the 2 × 2 camera array, while the narrowband receiving configuration, together with the optical filter and polarizer, improves the spectral selectivity of the coherent signal and suppresses unwanted background light. The four synchronously captured sub-aperture images

provide sparse Fourier-domain sampling of the target field and serve as the input to the proposed meteorology-aware computational restoration framework.

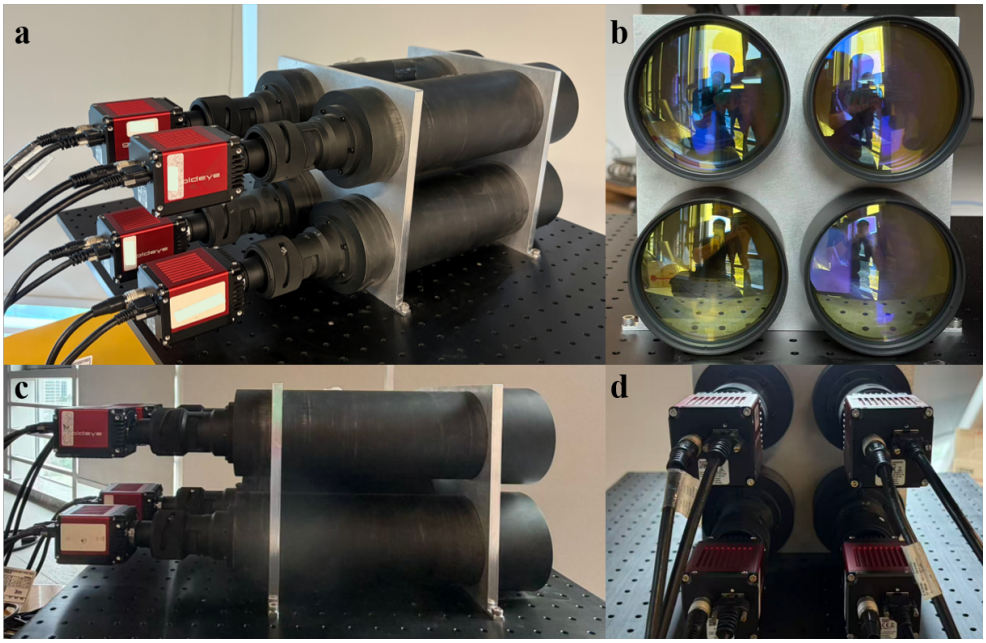


Fig. S1: **Compact receiving optics for the 2 × 2 camera-array system. a-d** Different views of the camera array. The compact customized receiving lens facilitates the integration and alignment of the 2 × 2 camera array for kilometer-scale single-shot FP imaging.

The main system-level advantage of this configuration is that it eliminates mechanical scanning during data acquisition. Therefore, the sub-aperture measurements are acquired under nearly the same atmospheric realization, which reduces temporal mismatch caused by atmospheric fluctuation, platform vibration, or target motion. This feature is particularly important for long-range imaging, where turbulence-induced phase aberration, beam wander, scintillation, and speckle may change with time. Compared with sequential scanning schemes, the proposed 2 × 2 single-shot configuration is more compatible with dynamic atmospheric environments and provides a practical route toward real-time or near-real-time far-field FP imaging.

Furthermore, the single-shot hardware design is naturally compatible with meteorology-aware computational restoration. In our experiments, the sub-aperture images were acquired over kilometer-scale propagation distances while meteorological parameters, including the refractive index structure constant, humidity, temperature, visibility, and wind speed, were recorded synchronously. This design enables the imaging data and environmental measurements to be paired in a physically meaningful manner, providing a more realistic basis for modeling and mitigating atmosphere-induced degradation in far-field coherent imaging.

Overall, the proposed system represents a transition from mechanically scanned macroscopic FP toward camera-array-based single-shot macroscopic FP at the kilometer scale. By combining simultaneous multi-aperture acquisition, active coherent illumination, and meteorology-aware computational restoration, the system improves acquisition efficiency, reduces temporal inconsistency among sub-apertures, and enhances the practicality of macroscopic FP for long-range imaging under dynamic atmospheric conditions.

1.2 Acquisition and calibration

We conducted sub-aperture image acquisition experiments in the tunnel at imaging distances ranging from 600 m to 1000 m. At each distance, the laser power used differed: as the imaging distance increased, we appropriately increased the laser power to ensure that, under the same 30 ms exposure time and 0 camera gain, the images acquired at all distances achieved suitable exposure control. The remaining image acquisition and preprocessing procedures were kept consistent across all distances. The 2×2 camera array was controlled by image acquisition software; after a one-click trigger of the image acquisition signal, the four-channel signals were transmitted through their respective Gigabit Ethernet ports, and the four-channel images ultimately acquired had identical timestamps.

For the acquired sub-aperture images, we first perform global denoising, i.e., subtracting the corresponding background noise (with the laser power set to 0, 10 sets of four-channel images were captured, and the pixel mean was calculated to obtain the background noise). Then, we roughly crop the images of each channel to obtain the per-channel images to be registered. The four images are registered by using a phase-correlation-based registration method. After selecting the first image as the reference image, the cross-power spectrum of any one of the remaining three images and the reference image is:

$$C\left(u,v\right)=\frac{F_1^*\left(u,v\right)F_i\left(u,v\right)}{\left|F_1(u,v)\right|\left|F_i(u,v)\right|}=e^{-j2\pi(\frac{ux_0}{M}+\frac{vy_0}{N})} \tag{1}$$

where $F_1(u,v)$ is the Fourier transform of the reference image, $F_1^*\left(u,v\right)$ is the complex conjugate of $F_1(u,v)$, $F_i(u,v)$ is the Fourier transform of one of the remaining three images, M and N are the dimensions of the image after roughly cropped, x_0 is the horizontal displacement, and y_0 is the vertical displacement.

Applying the inverse Fourier transform to the cross-power spectrum yields:

$$c\left(x,y\right)=\frac{1}{MN}\frac{sin[\pi(x-x_0)]}{sin[\frac{\pi}{M}(x-x_0)]}\frac{sin[\pi(y-y_0)]}{sin[\frac{\pi}{N}(y-y_0)]} \tag{2}$$

Then $c\left(x,y\right)$ can be approximated as a two-dimensional impulse function, and the horizontal displacement x_0 and vertical displacement y_0 of the image can be obtained by locating the peak of the two-dimensional impulse.

The registered images are then finely cropped to ensure that all valid information is fed into the model while reducing the memory occupied by invalid information. Other response differences are not handled separately in this system; instead, they are fed into the network together with environment-induced degradation for iterative optimization. After the training is completed, the reconstruction performance is tested following the

same preprocessing procedure. Since the iterative training incorporates the learning of the residual response differences among channels, the model can progressively correct the response differences among channels during weight computation at the test stage, ultimately reconstructing the intensity image. This acquisition-calibration-training-reconstruction pipeline can mitigate the limitations imposed by system fabrication on imaging quality and offer high repeatability.

2 Additional Experiments

All experiments were conducted in the kilometer-scale tunnel equipped with relatively comprehensive atmospheric turbulence monitoring. Sub-aperture images were acquired at propagation distances of 600 m, 800 m, and 1000 m, and meteorological parameters including refractive index structure constant, humidity, temperature, visibility and wind speed were measured in the kilometer-scale tunnel at 10-second intervals, enabling a systematic assessment of imaging performance across extended ranges. The details of our experimental setup in this section was the same as Table 1 presented in the main text.

2.1 Comparison of Different Distances

To further substantiate the efficacy of our method, we collected real-world data by the 2 $\times$ 2 camera array with the imaging distance ranging from 600 m to 1000 m. As shown in Fig. S2, the conventional FP algorithm was unable to reconstruct better images than the single-shot images for the structured frame with some circular holes, while the other four algorithms could reconstruct the outline of the frame. In addition, MPGNet fulfilled the best circular holes with the imaging distance ranging from 600 m to 1000 m, other algorithms with some degree of deformation, especially within the red rectangular box of Fig. S2. The reconstruction results in Fig. S2 manifested that

our MPGNet still outperformed other methods under varying realistic environments with different distances, demonstrating the robustness and efficiency of our approach.

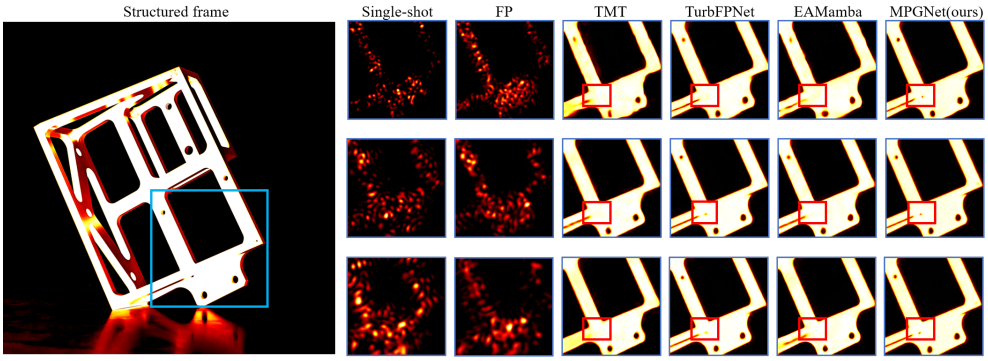


Fig. S2: **Comparison with the imaging distance ranging from 600 m to 1000 m for the structured frame with some circular holes.** The 1st row was one of the single-shot images obtained by the 2×2 camera array at a imaging distance of roughly 600 m and the corresponding reconstruction results listed adjacently for better evaluation. The imaging of the 2nd row took place at a distance of roughly 800 m, the 3rd row up to 1000 m. The input data of all the algorithms were the real single-shot images.

2.2 Comparison of Different Methods

Since the scarcity of training and testing coherent data with relatively comprehensive atmospheric turbulence monitoring parameters such as humidity, visibility, refractive index structure constant and other factors, we address the evidence of improved structural discrimination and reconstruction quality under the tested conditions facilitated by our data synthesis method. The details of our experimental setup in this section was the same as Section 2.1 and more diverse scenes were reported in Figs. S3, S4 and S5.

We employed the conventional FP algorithm [1], TMT [2], TurbFPNet [3] and EAMamba [4] as the baseline algorithms to evaluate the performance of our proposed algorithm on the benchmark datasets including DIV2K [5] and Flickr2K [6]. Because

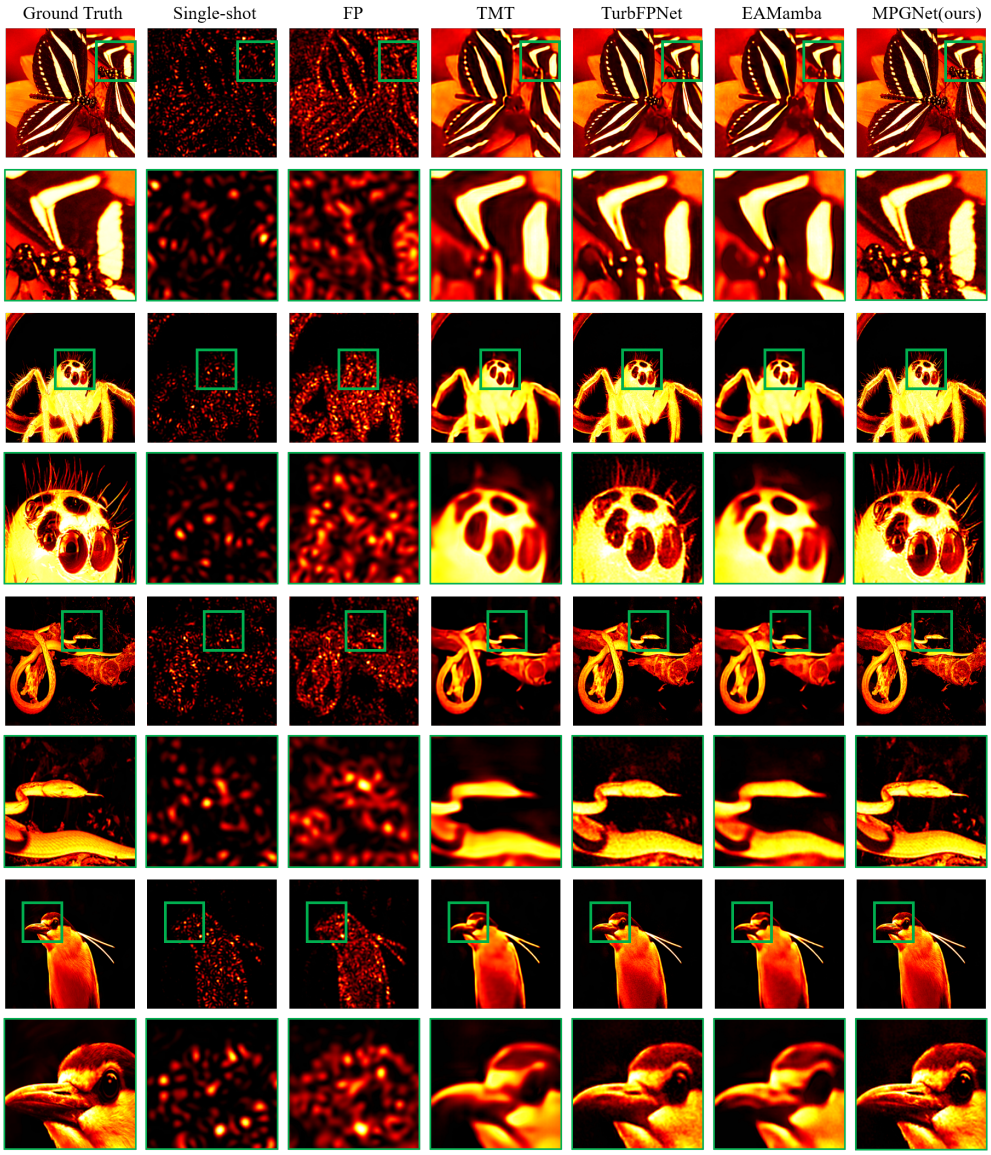


Fig. S3: **Visual comparisons of MPGNet with recent baselines on animals of the benchmark datasets.** Original images from the benchmark datasets were listed in the 1st column as ground truths. The 2nd column was one of the simulated single-shot images produced by the degradation module. Reconstruction results were listed in other columns for better evaluation and the input data of all the algorithms were the simulated single-shot images.

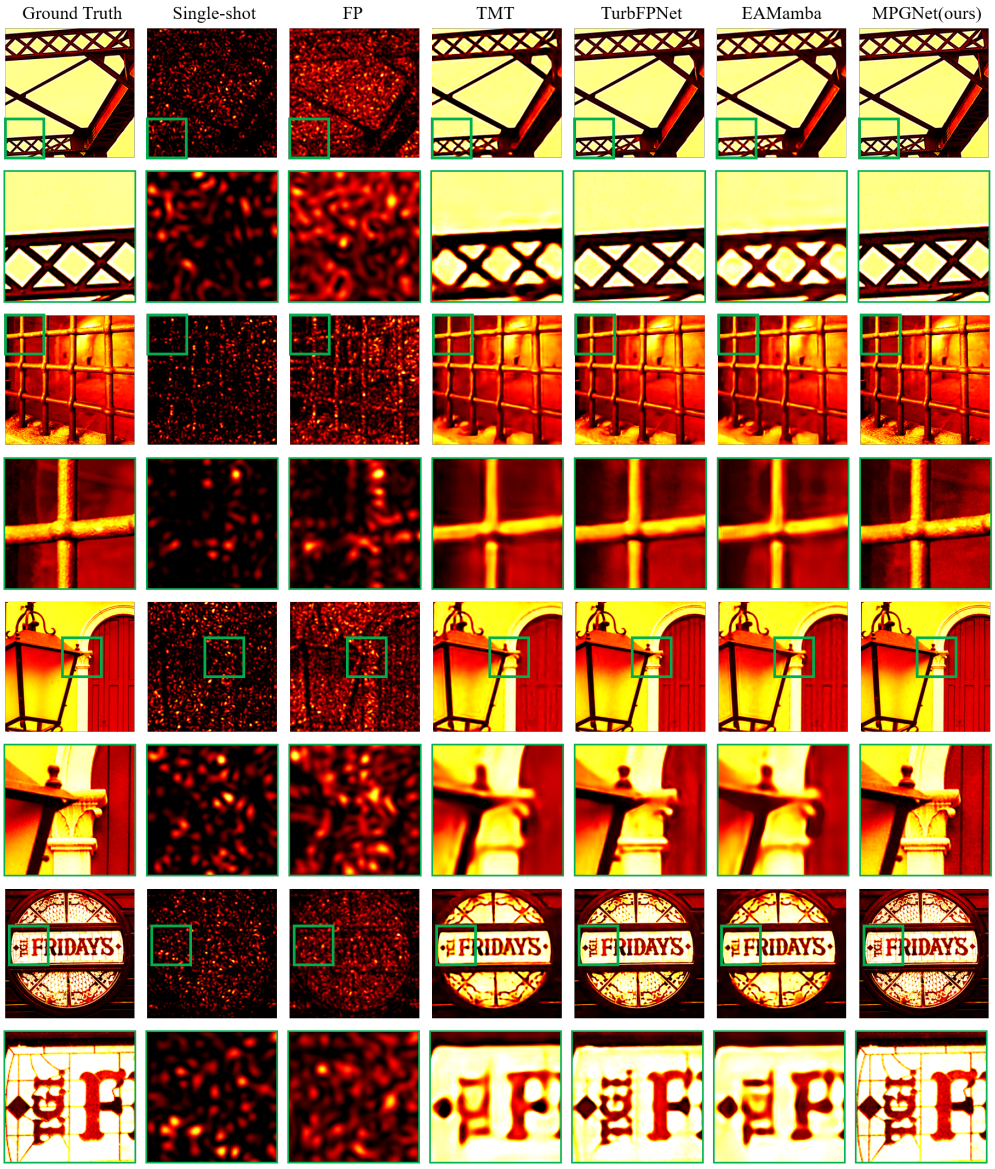


Fig. S4: **Visual comparisons of MPGNet with recent baselines on buildings of the benchmark datasets.** Original images from the benchmark datasets were listed in the 1st column as ground truths. The 2nd column was one of the simulated single-shot images produced by the degradation module. Reconstruction results were listed in other columns for better evaluation and the input data of all the algorithms were the simulated single-shot images.

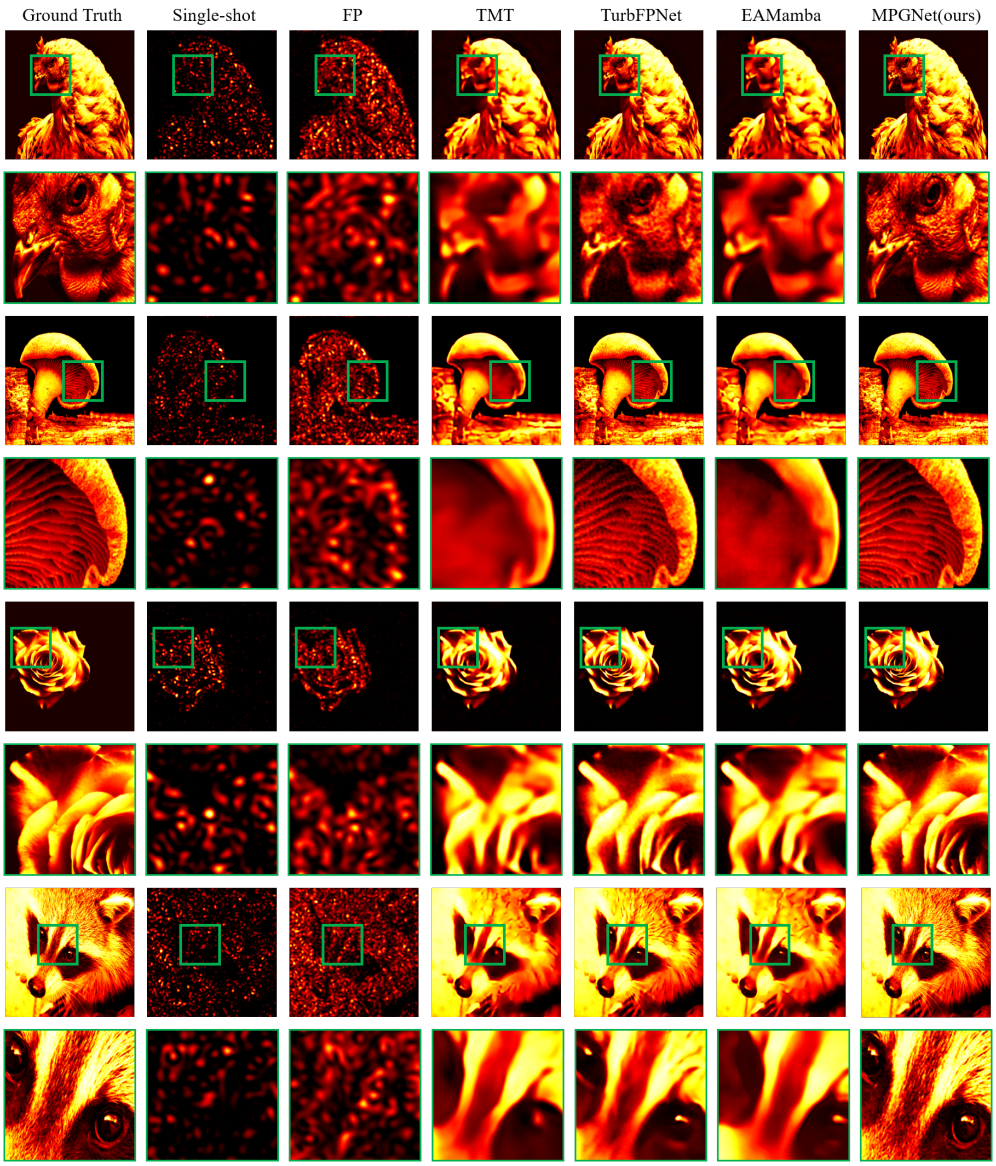


Fig. S5: **Visual comparisons of MPGNet with recent baselines on other static scenes of the benchmark datasets.** Original images from the benchmark datasets were listed in the 1st column as ground truths. The 2nd column was one of the simulated single-shot images produced by the degradation module. Reconstruction results were listed in other columns for better evaluation and the input data of all the algorithms were the simulated single-shot images.

our input data had only four sub-aperture images, not satisfying the conditions of stable convergence for traditional FP methods, the conventional FP algorithm was unable to reconstruct better images than the original sub-aperture images for all samples. It has been worth mentioning that our model was able to produce more conservative results without hallucinating many nonexistent details, delivering more reliable results than other models, as shown in Figs. S3, S4 and S5. Therefore, our MPGNet has still held the best generalizability among competing algorithms under the tested conditions.

2.3 Effects of Complementary Aperture Information

To validate the contribution of complementary aperture information, we compared three configurations of the network inputs: (a) one aperture; (b) one aperture repeated across four channels; (c) four distinct apertures, our original model. As shown in Fig. S6e, feeding the network with one aperture repeated across four channels yielded nearly identical final outputs compared with directly inputting the single-channel image, since the total effective input information remains unchanged.

Although the reconstruction result obtained by inputting only a single-aperture image (or by replicating that image into a four-channel input) could resolve the minimum distinguished element at (Group -1, Element 1), the edge textures of the bars exhibited a certain degree of distortion, and hallucination effects have already appeared at (Group -2, Element 4) and (Group -2, Element 5), where three alternating high-contrast bars arranged horizontally and vertically were partially connected or merged in some regions. Therefore, stable discrimination of the (Group -2, Element 4) and (Group -2, Element 5) elements could not be achieved. In contrast, when the 2×2 sub-aperture images were used as input, the reconstruction result could resolve the minimum distinguished element at (Group -1, Element 6), and the edge textures of the bars are clear and accurate. Moreover, the four-aperture input improved chart-element

discrimination by approximately 1.78-fold relative to the single-aperture reconstruction under the tested conditions. This improvement factor was also related to the level of interference in the imaging environment, and therefore did not exactly correspond to the ideal 2.1-fold synthetic aperture size.

These results have demonstrated that images from multiple apertures provide effective complementary information for long-range image reconstruction, which can not only counteract image distortion caused by a certain degree of environmental interference but also improve the image resolution capability.

2.4 Mismatch-conditioned Evaluation

To further validate the generalizability of our proposed model, we conducted experiments on unseen distances and meteorological conditions.

As shown in Fig. S7, we retrained our proposed model as MPGNet-V1, which was trained only using the synthetic dataset corresponding to imaging distances of 600 m and 800 m and tested on the real dataset corresponding to imaging distance of 1000 m. Compared with our original reconstruction results, for the object of concentric circular stripes converging towards the center, the boundary of the innermost circle was clear in both cases, and the clarity of the outer multi-layer gradient rings was also comparable. For the fuel gauge, the boundary of the pointer in the MPGNet-V1 reconstruction appeared jagged, and the fuel tank icon partially adhered to the triangular icon on the left, while the remaining parts were comparable.

For the meteorology-conditioned datasets, we classified the synthetic dataset with a refractive index structure constant greater than $10^{-16}\ m^{-2/3}$, a wind speed greater than 5 m/s, or a visibility less than 10 km as the relatively severe environmental degradation dataset, and grouped the rest into the relatively mild environmental degradation dataset. We then retrained our model as MPGNet-V2, which was trained only

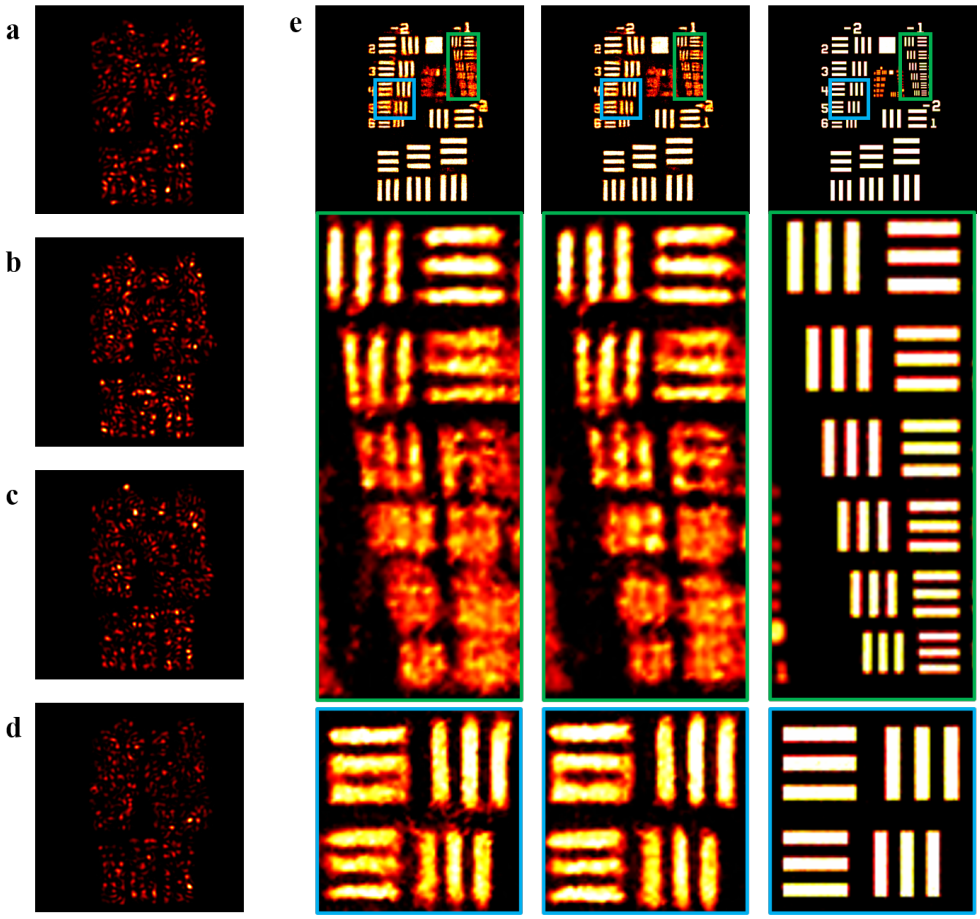


Fig. S6: **Comparison of reconstruction outputs of MPGNet under different input images. a-d** The experimentally-captured sub-aperture images at 1000 m. **e** The reconstructed result with only one aperture (we used panel **a**) as the input image was listed in the 1st column. The 2nd column was the reconstructed result with one aperture (we used panel **a**) repeated across four channels as the input images, and the 3rd column was the reconstructed result with four distinct apertures (we used panels **a-d**) as the input images. The corresponding zoom-in details were shown to the bottom of these results.

using the relatively mild environmental degradation dataset and tested on the relatively severe environmental degradation dataset. As shown in Fig. S8, for the resolution chart, the refractive index structure constant recorded in the YAML file was 8.62 ×

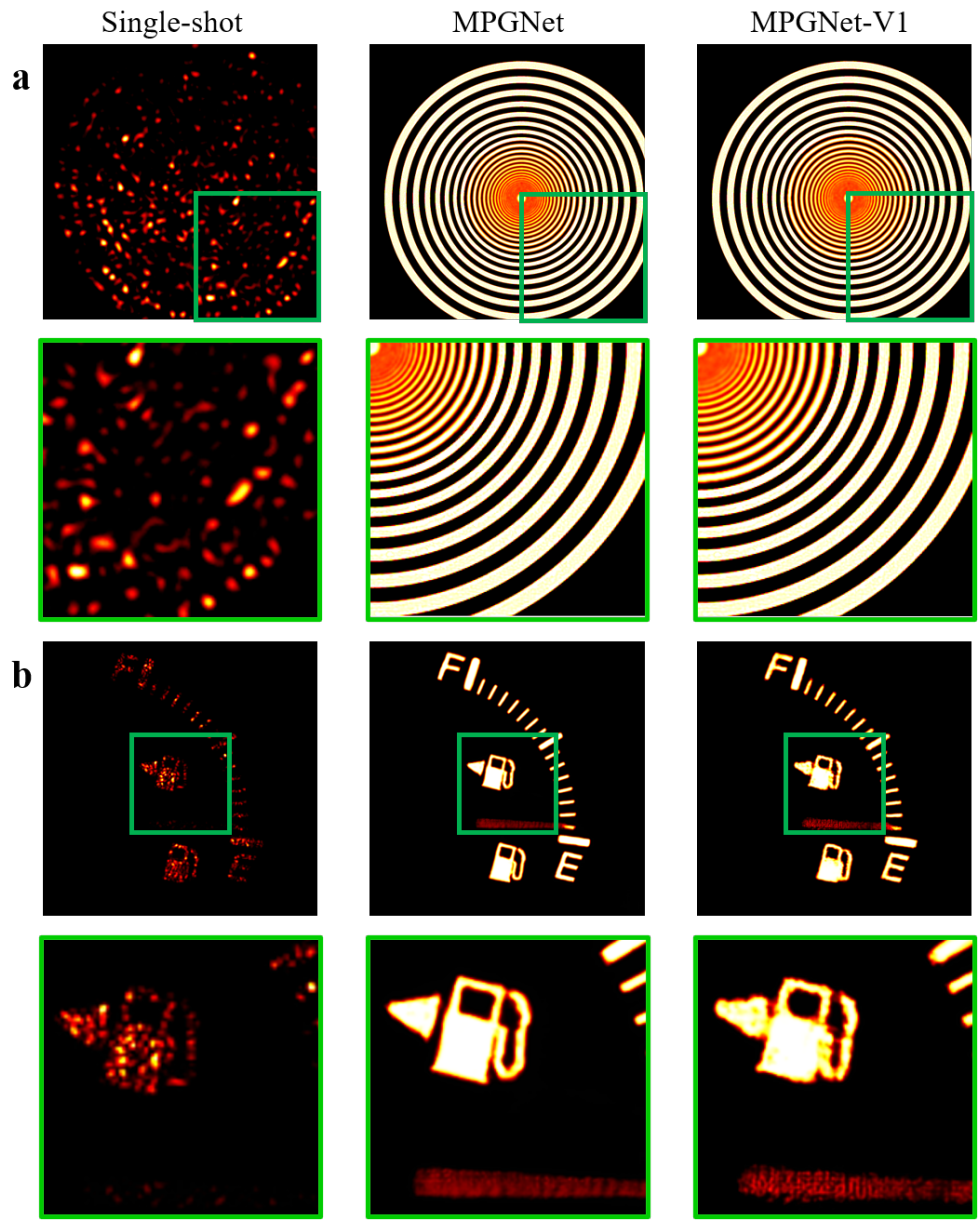


Fig. S7: **Comparison of reconstruction outputs of MPGNet with different distance-conditioned training datasets. a** Concentric circular stripes converging towards the center. **b** Fuel level gauge. The experimentally-captured sub-aperture images at 1000 m were listed in the 1st column. The 2nd column was the reconstructed results of our original MPGNet, and the 3rd column was the reconstructed results of MPGNet-V1, which was trained on two distances and tested on the held-out third. The corresponding zoom-in details were shown to the bottom of these results.

10^{-14} $m^{-2/3}$, which was relatively far from the meteorological parameters MPGNet-V2 was trained on. Consequently, distortions appeared in multiple elements; even the horizontal bars of (Group -2, Element 1) showed partial adhesion. For the annular fringes of variable width, the visibility recorded in the YAML file was 9.23 km, which was relatively close to the meteorological parameters MPGNet-V2 was trained on. Therefore, the resolving capability of the reconstructed annular fringes was comparable to that of our MPGNet while there was staining on the outermost and innermost fringes, which was attributed to interference from different meteorological conditions.

We could observe that although training on two distances and testing on the held-out third remained some meteorological condition differences, they were relatively minor and had little effect on the final reconstruction results. However, even if the training data included 1000 m, large differences in meteorological conditions would still lead to incompletely corrected blurring, distortion, and other issues in the reconstructed results.

To reduce model mismatch scenarios, we can use our pre-trained model as the teacher model and employ techniques such as distillation learning and transfer learning to obtain a more suitable model on incrementally expanded datasets, which cover more distances, more meteorological conditions and new scenarios.

3 Appendix

3.1 Meteorological Sampling

By analyzing the statistical patterns of the meteorological data recorded in the tunnel, we find that these meteorological data fluctuate only slightly over short periods, as shown in Fig. S9. We therefore consider that the values recorded by the 10-second sampling lie close to the true meteorological data corresponding to the images we recorded. In addition, our model undergoes two rounds of training. In the first round, the simulated datasets and their corresponding YAML files are used directly. In the

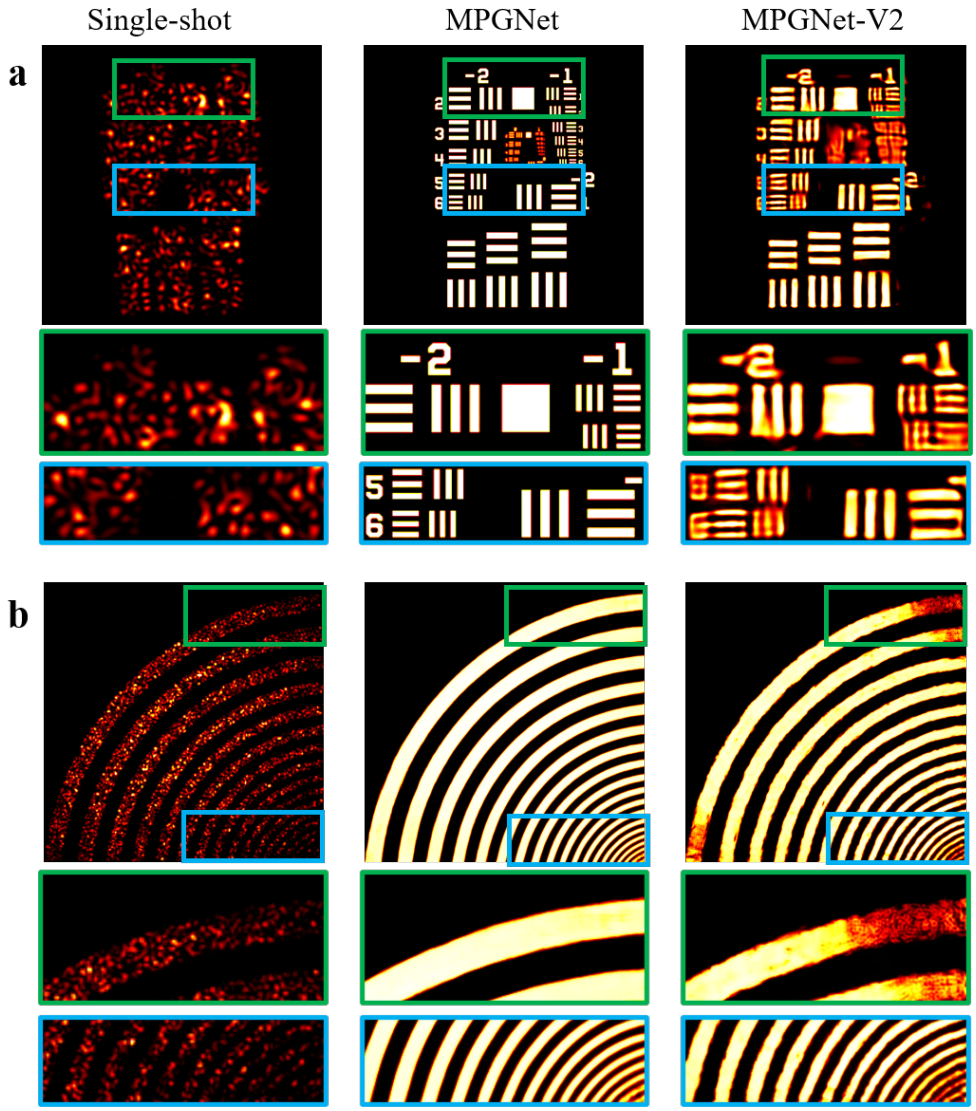


Fig. S8: **Comparison of reconstruction outputs of MPGNet under different meteorology-conditioned training datasets. a** The resolution chart. **b** The annular fringes of variable width. The experimentally-captured sub-aperture images at 1000 m were listed in the 1st column. The 2nd column was the reconstructed results of our original MPGNet, and the 3rd column was the reconstructed results of MPGNet-V2, which was trained on the relatively weak turbulence and tested on the relatively strong one. The corresponding zoom-in details were shown to the bottom of these results.

second round, the result of the first round is used as a pretrained model, while the meteorological data recorded in the YAML file are randomly perturbed around the original values by no more than 10% to serve as the new meteorological input to the network, and network fine-tuning training continues using the same simulated images as in the first round. This training scheme allows for a certain instrument-precision error in the meteorological data while enhancing the generalization of the network.

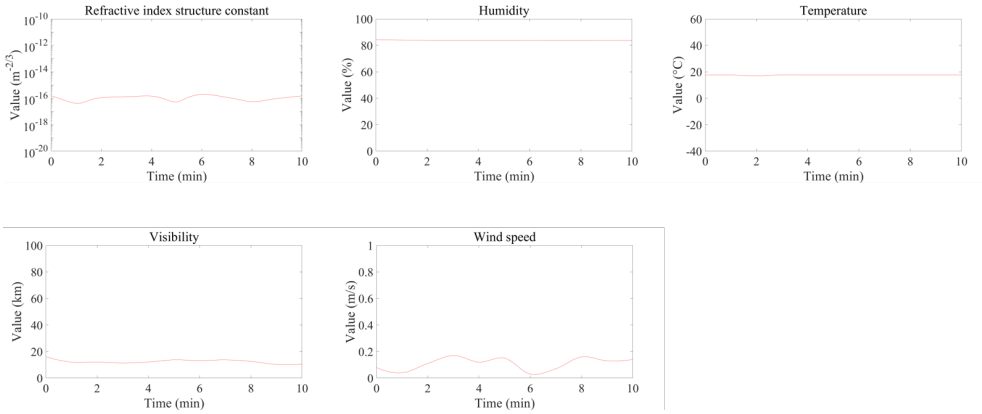


Fig. S9: **Time series of five meteorological parameters including refractive index structure constant, humidity, temperature, visibility and wind speed, with corresponding recording times from 10:00 to 10:10 on April 15, 2026.** The data was provided by Temporal-spatial manipulation Infrastructure for vector Fields in Optics-Test Facility (TIFO-TF).

3.2 The Customized Resolution Chart

Apart from accurate positioning and quantity of the three vertical and three horizontal alternating high-contrast bars, we adopt a Michelson contrast threshold of 0.15 as an operational criterion for bar resolvability. The contrast is defined as:

$$C = \frac{I_{max} - I_{min}}{I_{max} + I_{min}} \tag{3}$$

where I_{max} is the maximum value of the intensity of the line pair, I_{min} is the minimum value of the intensity of the line pair.

Therefore, the reconstruction results of TMT, EAMamba, TurbFPNet and MPGNet exhibited the minimum distinguished element at (Group -2, Element 3), (Group -2, Element 4), (Group -1, Element 4) and (Group -1, Element 6), respectively, as the corresponding intensity profiles shown in Fig. 4 of the main text. The corresponding physical sizes were shown in Fig. S10. Besides, the physical size of (Group -3, Element 4), shown in Fig. S10a, corresponded to the widest element of the customized resolution chart that could not be resolved in the sub-aperture images mentioned in the main text.

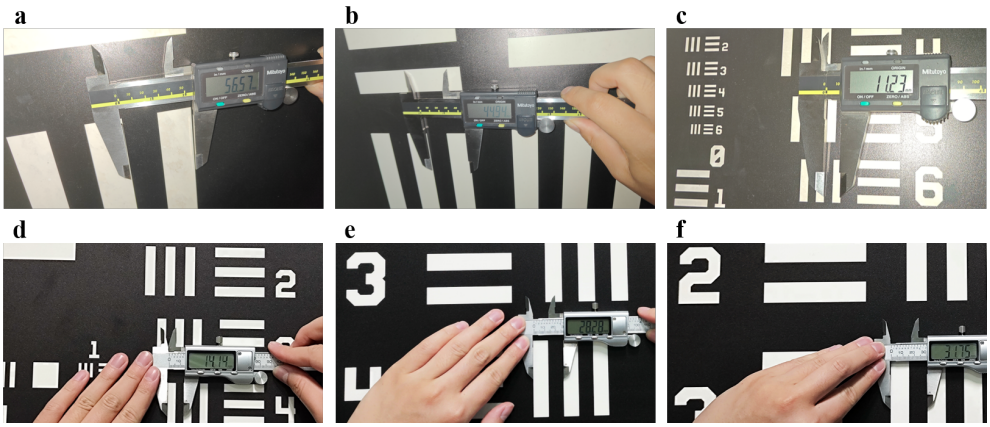


Fig. S10: **The physical scale of the customized resolution chart. a** (Group -3, Element 4), corresponding to a physical size of 56.57 mm. **b** (Group -3, Element 6), corresponding to a physical size of 44.94 mm. **c** (Group -1, Element 6), corresponding to a physical size of 11.23 mm. **d** (Group -1, Element 4), corresponding to a physical size of 14.14 mm. **e** (Group -2, Element 4), corresponding to a physical size of 28.28 mm. **f** (Group -2, Element 3), corresponding to a physical size of 31.75 mm.

The customized resolution chart was fabricated according to the USAF standard; the measurement accuracy of the line-pair width might be approximately 0.01 mm, which could be attributed to fabrication tolerances or to the precision error introduced by averaging multiple measurements acquired with a vernier caliper.

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